

Solution to the challenge proposed by Doppel

An information-theoretic answer and an exact Boolean-network study

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Abstract

Doppel’s phishing challenge reached by means **3Blue1Brown**, asks which phrase frequencies maximise expected payoff while remaining within a divergence-based alarm budget. The optimum is an exponential reweighting of normal traffic, with its strength set by the budget. I give the proof and boundary cases, then examine a constrained version in which a Boolean network generates the distribution and the attacker may change one connection. A ten-node example shows how the best admissible payoff compares with the unrestricted bound. A verified study of 106,789 networks supports the computational construction. In the spirit of Kolmogorov’s algorithmic complexity, I use the length of an exact generative programme as a secondary description of the mechanism.

1 Introduction

The challenge asks for an optimal distribution, given normal phrase frequencies, phrase payoffs and a KL-divergence alarm rule [1]. I solve this distributional problem analytically, including its boundary cases. A Boolean-network experiment then evaluates the choices attainable through specified structural changes, with exact enumeration at the studied sizes. Programme length provides a secondary description of the generating rules. These contributions distinguish the optimal frequency mixture, the choices available to a constrained mechanism and the size of its encoded representation. Supplementary material provides the full implementation and experimental evidence.

2 The challenge and its solution

A phrase is a particular wording; a phrase category groups wordings with a shared function. For example, *reply quickly* and *respond today* could both express urgency. A message may contain phrases from several categories, such as an offering together with urgency. The original puzzle assigns probabilities to individual phrases. Our Boolean-network extension instead represents a message by the presence or absence of category features. Using the order offering, request, urgency, 101 denotes an offering and urgency without a request, regardless of the particular wordings used.

I treat each complete combination as one message type. Its probability describes those features occurring together: the distribution over types is a joint distribution. One output is one combination, not a distribution by itself. This extends the alphabet of the original puzzle while retaining its mathematical form. Below, outcome i denotes a phrase in the puzzle or a complete combination in the extension. Let q_i be its normal frequency, p_i its chosen frequency and $s_i \in [0, 1]$ its specified success probability when unflagged. A campaign is the collection of messages evaluated together: $p_i = 0.2$ means that outcome i is selected in 20% of its messages

on average. Probabilities of individual features need not sum to one, since features can coexist; probabilities of the complete types do. The payoff belongs to the complete outcome, not to a sum of the phrases it contains. Multiplying p_i by s_i weights that outcome's success by how often it is used. Adding these contributions gives expected payoff per message. The detector flags a campaign when $D_{\text{KL}}(p\|q) > C$, where $C \geq 0$ is the alarm budget. Treating a flagged campaign as receiving zero payoff, the nonnegative payoffs and feasible baseline give the problem

$$\max_p \sum_i p_i s_i \quad \text{subject to} \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad D_{\text{KL}}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} \leq C. \quad (1)$$

The maximum asks for the best frequency mixture. Nonnegative entries summing to one make it a probability distribution. In KL divergence, p_i/q_i compares a chosen frequency with normal traffic; the logarithm of this ratio is averaged using p . The result is zero at $p = q$ and positive otherwise. Thus the alarm limits a shift in frequencies, rather than the success rate itself. Natural logarithms express C in nats. If $q_i = 0$, using that outcome with positive probability gives infinite KL. For finite C we therefore set $p_i = 0$ and restrict sums to $q_i > 0$. Terms with $p_i = 0$ contribute zero. The puzzle uses exact distributions; estimating them from a finite batch is a separate observation problem.

2.1 The answer

Reweight each baseline frequency by an exponential factor:

$$p_i(\beta) = \frac{q_i e^{\beta s_i}}{Z(\beta)}, \quad Z(\beta) = \sum_j q_j e^{\beta s_j}. \quad (2)$$

First multiply each normal frequency by $e^{\beta s_i}$, the exponential of βs_i . A larger payoff receives a larger multiplier. Then add all these weights to obtain $Z(\beta)$ and divide each weight by that total. This normalisation makes the new frequencies sum to one. The parameter β controls how strongly payoffs affect the mixture: at zero all multipliers are one and $p = q$. In the interior case, increase β until $D_{\text{KL}}(p(\beta)\|q) = C$, using the available alarm budget. The boundary cases below specify when this applies.

For an illustrative example, let $q = (0.5, 0.3, 0.2)$, $s = (0.1, 0.4, 0.9)$ and $C = 0.25$. Then $\beta \approx 2.09644$ and $p \approx (0.23444, 0.26383, 0.50173)$. In 1,000 selections, the expected outcome counts change from (500, 300, 200) to approximately (234, 264, 502). The highest-payoff outcome becomes more frequent, while the mixture remains at the alarm limit. Expected payoff rises from 0.35 to 0.580533, calculated by multiplying each new frequency by its payoff and adding. Appendix A gives further detail.

2.2 Why it is optimal

Write $\mathbb{E}_p[s] = \sum_i p_i s_i$ for expected payoff under p . To prove optimality, compare any feasible mixture with $p(\beta)$. KL divergence between them cannot be negative, so for $\beta > 0$,

$$0 \leq D_{\text{KL}}(p\|p(\beta)) = D_{\text{KL}}(p\|q) - \beta \mathbb{E}_p[s] + \log Z(\beta), \quad (3)$$

$$\mathbb{E}_p[s] \leq \frac{C + \log Z(\beta)}{\beta}. \quad (4)$$

The first line rewrites that nonnegative difference in terms of payoff and divergence from normal traffic. Replacing the latter by its allowed maximum C gives the second line: a ceiling on every feasible payoff. When $p = p(\beta)$ and its divergence equals C , both inequalities become equalities. The proposed mixture reaches the ceiling and is optimal. To find its weight, observe that

$$\frac{d}{d\beta} D_{\text{KL}}(p(\beta)\|q) = \beta \text{Var}_{p(\beta)}(s). \quad (5)$$

The derivative describes how divergence changes as β increases. The variance measures the spread of payoffs under the current mixture. If the supported payoffs differ, this spread is positive for finite β , so divergence increases strictly for $\beta > 0$. We can therefore find the unique interior weight by adjusting one number until the divergence reaches C .

2.3 Boundary cases

At $C = 0$, the only feasible distribution is q . If all supported payoffs are equal, every feasible distribution is optimal. Otherwise, let S_* contain the supported outcomes with largest payoff and let $Q_* = \sum_{i \in S_*} q_i$, their combined frequency in normal traffic. Within this set, $q_{S_*}(i) = q_i/Q_*$ preserves their relative normal frequencies and rescales them to sum to one. For a mixture using only these best outcomes,

$$D_{\text{KL}}(p||q) = D_{\text{KL}}(p||q_{S_*}) - \log Q_* \geq -\log Q_*.$$

The remaining KL term cannot be negative. Preserving those relative frequencies makes it zero, so $-\log Q_*$ is the smallest budget permitting maximum payoff. For $C \geq -\log Q_*$, choose $p_i = q_i/Q_*$ on S_* and zero elsewhere. For $0 < C < -\log Q_*$, the finite positive β above attains the optimum. A deterministic choice of a maximum-payoff outcome k is also optimal when $C \geq -\log q_k$: this is the divergence of using that outcome every time. For the example above, using only outcome 3 requires $-\log(0.2) \approx 1.609$ nats, exceeding the budget of 0.25. Mixing is required in the interior case, but need not be required at larger budgets.

3 When a mechanism restricts the choice

The unrestricted solution permits any feasible probability vector. To examine a structural restriction, let a Boolean network update its N binary nodes synchronously by $x_{t+1} = F_A(x_t)$. Here x_t is the current vector of zeros and ones; F_A applies the local gates using connections recorded in A to produce the next vector. The one-step repertoire lists every possible input vector and its output. An input is a generator state, rather than text submitted to a detector. A permitted attack adds or removes one connection while keeping gate types and parameters fixed. It changes an internal dependency and can therefore change which states occur and how frequently.

Each node now represents one category feature and each complete output one message type, as defined in Section 2. The gates model logical dependencies in how combinations are generated. An AND rule, for example, activates an output feature only when all its connected input features are present; XOR activates it when an odd number are present. Shared inputs and such rules can make output features dependent. This gives an abstract model of rule-based composition, rather than independently choosing every feature. The experimental rules and payoffs are declared mathematical choices, without calibration to text or recipient responses. The results therefore concern the generation and frequencies of abstract message types; they do not establish grammatical validity, intelligence or persuasive effectiveness.

Feeding each output back as the next input produces a trajectory. There are only 2^N states; once a state repeats, the deterministic rules repeat the same sequence, forming a cycle. Its basin contains every initial state that eventually reaches it. To define the observed distribution, start uniformly over all 2^N states and sample a phase uniformly from the eventual cycle. If state x lies on a cycle of length ℓ whose basin contains b initial states, then

$$\mu_A(x) = \frac{b}{2^N \ell}.$$

The fraction $b/2^N$ is the chance of reaching that cycle under uniform initialisation. Dividing by ℓ shares its probability equally among its phases. Transient states receive zero long-run mass because their finite visits contribute a vanishing fraction of an indefinitely long trajectory; this does not imply rejection or failure. The detector uses the resulting frequencies of whole combinations, rather than sequence order. Counting outputs after one update with uniformly sampled inputs gives a different observation: how many starting states produce each combination immediately. Both distributions are joint across features; cycles determine which combinations recur and their long-run weights. Our distribution averages over initial states and cycle phases; one run eventually visits only its own cycle. Appendix B gives the full observation convention.

The baseline network A_0 generates $q = \mu_{A_0}$ and each permitted network A generates $p = \mu_A$. The symbol μ denotes the frequencies calculated from that network’s cycles and basins; q remains fixed throughout the comparison. For a finite catalogue $\mathcal{A}$ of nonidentity changes—actual modifications, excluding the unchanged network—select

$$\max_{A \in \mathcal{A}: D_{\text{KL}}(\mu_A \| \mu_{A_0}) \leq C} \sum_x \mu_A(x) s(x).$$

For every candidate, compute its distribution, discard it if its KL exceeds C , and average its state payoffs $s(x)$ using the remaining frequencies $\mu_A(x)$. The maximum selects the highest such average. This is the original objective evaluated on a restricted set of choices. If none is feasible, report the result as unavailable; taking no action is excluded from this catalogue and can be added explicitly if desired. Its optimum cannot exceed the unrestricted optimum because every admissible network distribution is also a candidate in the unrestricted problem.

3.1 A ten-node example

Figure 1 shows the saved baseline `ring_n10_s1000`. Its 1,024 initial states lead to one fixed point and three eight-state cycles, with basin sizes 6, 148, 389 and 481. For example, each state of the second cycle receives mass $148/(1024 \cdot 8) = 37/2048$; the fixed point 0 receives $6/1024$. We obtain these counts by following the transitions from every initial state and grouping those that reach the same cycle. The 148 starting states give their cycle probability $148/1024$; its eight combinations share that weight equally. There are 432 distinct one-step outputs but only 25 states with positive long-run mass. Thus selecting the observation scheme changes which message types receive probability. A structural change can alter both cycle membership and basin sizes, changing those probabilities even when some recurring combinations are retained. The cycles, selected transitions and exact probability table appear in Appendix C.

There are 74 permitted edge additions. At $C = 0.25$, 13 are feasible; 45 have infinite forward KL because they introduce positive probability outside the baseline support. For payoff equal to the fraction of active nodes, the best change adds source 4 to target 8. It achieves $51/64 = 0.796875$, with KL about 0.184507 nats. The saved unrestricted upper bound is 0.847502. For payoff equal to one at state 0 and zero elsewhere, the best attainable value remains $6/1024 = 0.005859375$ at all four evaluated budgets (Figure 2).

The target payoff counts how often one specified message type appears. The active-node payoff averages the proportion of features present; it does not reward additional interactions between them. In Figure 2, the catalogue curve selects the best candidate at each budget, so it can stay flat when extra budget admits no better choice. The stored upper bounds evaluate the ceiling in Section 2.2 over a fixed grid of β values, with explicit boundary cases. They need not equal the exact unrestricted optimum; displayed gaps may include numerical slack as well as structural restriction.

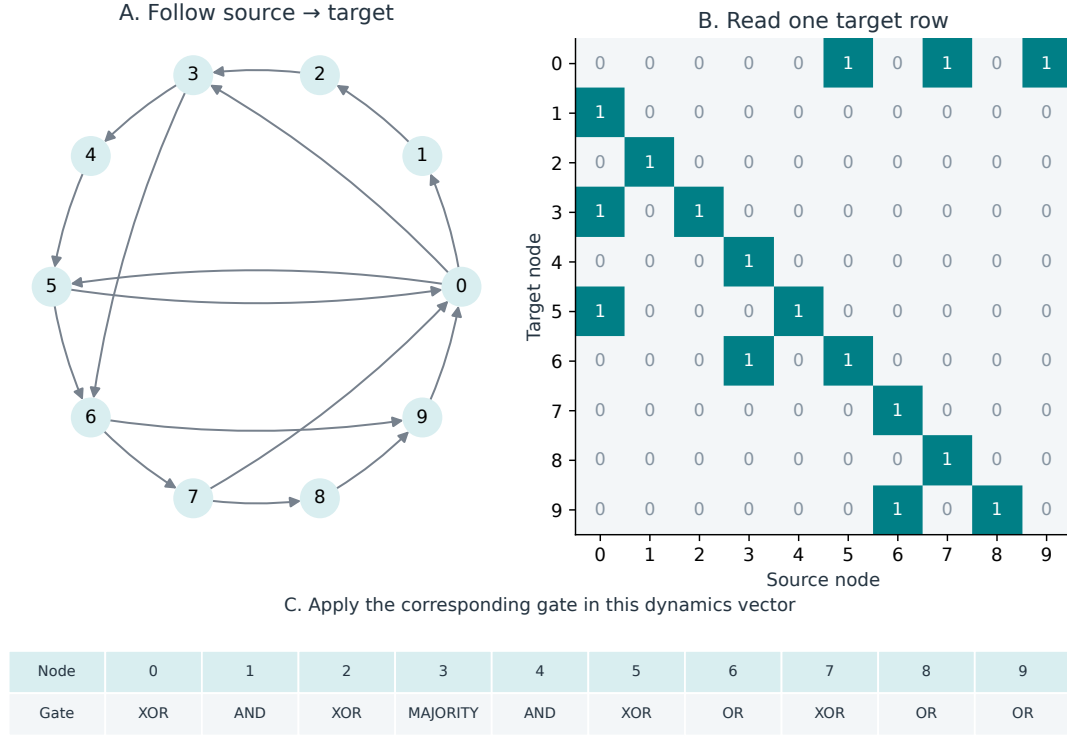


Figure 1: The worked generator: connections, adjacency matrix and local rules. An arrow feeds the target’s gate; all nodes update together. For example, node 0 computes the XOR of nodes 5, 7 and 9.

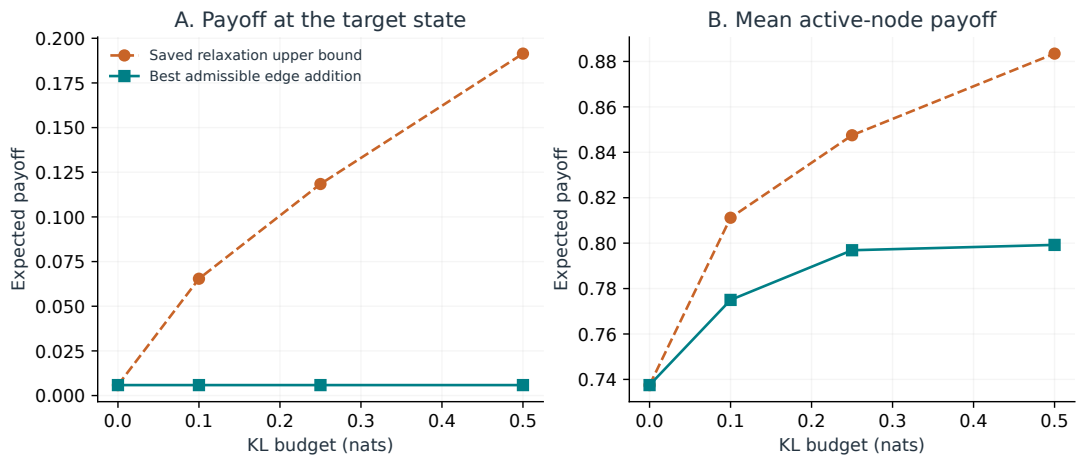


Figure 2: Best admissible payoff and saved unrestricted upper bound for the same catalogue. Markers are the four evaluated budgets; connecting lines guide the eye. Panel A rewards state 0; Panel B rewards the fraction of active nodes.

3.2 Verification and the role of complexity

The final study covers 106,789 distinct networks, 2,400 catalogues and 105,642 changes across four families and $N = 8, 10, 12$, with maximum indegree five. Exact transition reconstruction, cycles, basin weights and distributions passed the declared checks. At $C = 0.25$, no nonidentity change was feasible in 14 of 1,200 addition catalogues and 97 of 1,200 removal catalogues. Protocols and verification are in Appendix C; complete catalogue summaries are in Appendix D.

An executable decision graph [4] reproduces the worked network’s 10,240-bit transition table in 325 logical bits. It stores the logical dependencies that generate the feature combinations and their transitions. Executing it recovers every update used to calculate cycles and distributions. Its length, called causal description length under this encoding, measures the encoded rule structure of the complete mechanism. This motivates the connection to Kolmogorov complexity [3]: an executable description gives an upper bound on the shortest programme for the same transition table, including fixed interpreter overhead. Equal lengths need not identify equal behaviour: feasible 339-bit programmes yield linear payoffs from 0.737500 to 0.796875. We obtain that range by grouping the feasible changes by their encoded length and comparing their payoffs. Equal lengths can encode different rules, so length alone cannot determine payoff in this example. Catalogue selection uses the calculated distributions and payoffs. The BDM comparison [5] and full encoding are retained in the supplement.

4 Conclusion

Exponential reweighting solves the stated distributional challenge, with the alarm budget determining how far the attacker can favour higher-payoff outcomes. The network extension evaluates the same objective over joint feature distributions produced by permitted changes to a generator. Cycles and basin weights determine those distributions, allowing exact selection of the best admissible payoff within each catalogue. The executable programme preserves the generating rules, while its length provides a summary that need not determine payoff. The solution, constrained experiment and description-length analysis thus have distinct, explicit roles.

References

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Supplementary material

These sections provide the observation conventions, worked reconstruction, complete experimental context and secondary analyses supporting the main response. They introduce no additional network sample.

A Interpreting the distributional solution

The two distributions play different roles. The fixed reference q describes ordinary traffic; p describes the frequencies chosen by the attacker. Outcome i denotes a phrase in the puzzle or a complete message type in the network extension. Thus $q_i = 0.3$ means that outcome i accounts for 30% of baseline messages, while $p_i = 0.2$ means that the attacker uses it in 20% of messages on average. Neither symbol denotes the text of a message.

The ratio p_i/q_i compares the chosen frequency with the baseline. KL divergence averages the logarithm of this ratio under p : it is zero when $p = q$ and positive otherwise. The budget C therefore limits how far the attacker can shift the frequencies, as measured by this divergence. Natural logarithms express the budget in nats.

A phrase with $q_i = 0$ never occurs under the baseline model. Assigning it positive probability would place weight on an event that the baseline regards as impossible. The term $p_i \log(p_i/q_i)$ is then infinite, so such a choice cannot satisfy any finite budget. We must set $p_i = 0$ for these phrases and can restrict all sums below to $q_i > 0$. As usual, terms with $p_i = 0$ contribute zero to the divergence.

Here $e^{\beta s_i}$ is the exponential of βs_i : the baseline probability q_i is multiplied by this factor, not by βs_i itself. Larger payoffs receive larger factors, and $Z(\beta)$ makes the resulting probabilities sum to one. For two supported phrases, their relative frequencies satisfy

$$\frac{p_i(\beta)}{p_j(\beta)} = \frac{q_i}{q_j} e^{\beta(s_i - s_j)}.$$

Thus equal payoffs preserve the baseline ratio, while a higher payoff increases a phrase's weight relative to a lower-paying one.

Multiplication here changes a sampling weight, not the words inside a message. Its operational interpretation is to select outcomes with the new frequencies. For example, consider three abstract outcomes with $q = (0.5, 0.3, 0.2)$ and $s = (0.1, 0.4, 0.9)$. At $C = 0.25$, the solution has $\beta \approx 2.09644$ and $p \approx (0.23444, 0.26383, 0.50173)$. In 1,000 independent selections, the expected outcome counts change from (500, 300, 200) to approximately (234, 264, 502). Individual realised counts fluctuate. The puzzle applies its detector to the specified distribution; estimating frequencies from a finite batch would introduce an additional sampling problem.

The parameter β controls the strength of this preference. At $\beta = 0$ the formula gives $p = q$; increasing β shifts probability towards higher payoffs. It has no assigned physical meaning in this model, whose quantities are frequencies and success probabilities. For $\beta > 0$, its mathematical role is precise: the distribution in equation (2) maximises $\mathbb{E}_p[s] - D_{\text{KL}}(p||q)/\beta$. Hence $1/\beta$ is the penalty placed on divergence relative to payoff. In the interior case of Section 2.3, choose β so that $D_{\text{KL}}(p(\beta)||q) = C$.

B Network dynamics and observation model

B.1 Mapping the problem to a Boolean network

A Boolean network contains N nodes, each holding a zero or one. A local Boolean rule determines each node's next value from its connected inputs. All nodes update together. A network therefore defines a deterministic function $F_A : \{0, 1\}^N \rightarrow \{0, 1\}^N$, where A is its connectivity matrix and the local rules are held fixed.

The adjacency matrix records these connections: $A[\text{target}][\text{source}] = 1$ when the source feeds the target (Figure 1). We label each binary state by an integer address x , running from 0 to $2^N - 1$. Node i has weight 2^i , so node 0 is the least significant bit. In the code, its value is extracted as $(x \gg i) \& 1$. Printed bit strings follow node order, $x_0, \dots, x_{N-1}$.

The ordered input repertoire lists every state once, in address order. The corresponding output row contains the state after one update. These ordering conventions let the decoder construct the input addresses itself; a separate input table does not have to be transmitted.

B.2 Why cycles define the observed frequencies

The puzzle specifies p and q directly. A deterministic network instead specifies a rule for moving between states, so we must also say how it will be observed. Repeatedly feeding each output back as the next input produces a trajectory. There are finitely many states, so some state must eventually repeat. From that point onward the deterministic rule repeats the same sequence: a *cycle*. A fixed point is a cycle of length one. The states leading to a cycle form its *basin of attraction*.

To construct the distribution used in the challenge, start uniformly from all initial states, follow the trajectory to its cycle, and choose one position, or phase, uniformly on that cycle. If $\Gamma_A(x_0)$ is the cycle reached from x_0 , the resulting distribution is

$$\mu_A(x) = \frac{1}{2^N} \sum_{x_0 \in \{0,1\}^N} \frac{\mathbf{1}\{x \in \Gamma_A(x_0)\}}{|\Gamma_A(x_0)|}. \quad (6)$$

The factor $1/2^N$ gives each initial state equal weight. The indicator selects trajectories whose eventual cycle contains x , and the denominator divides that trajectory's weight among the cycle's states. Equivalently, if a cycle has b initial states in its basin and contains ℓ states, each state on it has probability

$$\mu_A(x) = \frac{b}{2^N \ell}.$$

Transient states have zero mass in this long-run distribution, even though they may occur during the initial part of a trajectory.

In the challenge analogy, a network is a finite-state generator and its state labels stand for complete message types. A cycle represents a repeated sequence of types after the generator settles. A larger basin makes that sequence more likely under the chosen initialisation; a longer cycle divides its mass among more types. This is an interpretation of the experimental generator, not a claim that natural text messages follow Boolean cycles. Uniform initialisation and uniform phase selection are explicit modelling choices. Figure 3 shows how they produce the distribution in the ten-node example.

Equation (6) also gives the time-averaged limiting distribution under uniform initialisation. It can differ from the distribution at one particular late update, because different runs may have synchronised phases. Counting output rows after just one update gives another distribution again. We use the long-run experiment consistently for both the baseline and each changed

network. The detector uses these marginal frequencies. It does not test the order in which cycle states occur; a detector sensitive to temporal patterns would pose a different problem.

Under the feature interpretation in the main text, μ_A is joint across the nodes and marginal with respect to time: it describes a whole feature configuration at one sampled phase, rather than a sequence of configurations. For example, the frequency of one feature alone is found by adding $\mu_A(x)$ over all states with that bit equal to one. Such individual feature frequencies do not in general determine μ_A . The detector here compares the complete state distributions. A detector using only individual feature frequencies would be a different observation model, even though both can be constructed from the same network.

B.3 What a perturbation changes

A *perturbation* is a specified change to the generator. Here it means adding or removing one permitted connection while keeping the gate types and parameters fixed. It changes which inputs a rule reads and can therefore change the transition table, the cycles and their basin sizes. The word *disturbance* is a general informal description; we use *perturbation* for these controlled edge changes.

The baseline network A_0 generates $q = \mu_{A_0}$. Each permitted network A generates a candidate $p = \mu_A$. The reference q stays fixed while we compare these candidates. An edge change is an abstract modification of the message generator’s internal dependencies. It does not represent a literal word replacement, and a single structural change need not produce a small change in message frequencies.

Perturbations let us ask which distributions the restricted generator can actually produce. The attacker chooses one of these changes; the detector then compares its resulting frequencies with the original baseline. This extends the unrestricted puzzle, in which the attacker may choose any feasible probability vector.

Challenge component	Network counterpart
Phrase t_i	A labelled state in the common alphabet $\{0, 1\}^N$
Ordinary distribution q	μ_{A_0} for the baseline network
Chosen distribution p	μ_A for an admissibly perturbed network
Phrase payoff s_i	A declared state payoff $s(x)$
Attacker’s choice	One admissible edge addition or removal
Judge or detector	The same rule: alarm if $D_{\text{KL}}(\mu_A \parallel \mu_{A_0}) > C$
Expected payoff	$\sum_x \mu_A(x) s(x)$

Table 1: What does each part of the challenge become? The distributions, payoff and alarm retain their mathematical roles. The attacker’s available choices acquire a structural restriction.

This construction preserves the distributions, payoff and alarm rule of equation (1), while restricting the available choices to a finite set. The state labels and payoffs are declared experimental quantities; they have not been fitted to human responses to text. The detector applies the supplied rule to exact distributions.

For a catalogue $\mathcal{A}$ of admissible nonidentity changes, define

$$V_{\mathcal{A}}(C) = \max_{A \in \mathcal{A}: D_{\text{KL}}(\mu_A \parallel \mu_{A_0}) \leq C} \mathbb{E}_{\mu_A}[s]. \quad (7)$$

If no change is feasible, the value is reported as unavailable. The baseline is retained for audit but excluded from this changed-network optimum. If taking no action is allowed, it can instead be included as an explicit additional choice. In either convention, the unrestricted solution provides

the upper bound

$$V_{\mathcal{A}}(C) \leq U(C), \quad U(C) = \inf_{\beta > 0} \frac{C + \log Z(\beta)}{\beta}, \quad (8)$$

with the boundary cases given above. This relation is the precise link between the original puzzle and the network experiment.

B.4 Description length as a secondary measurement

The network experiment also asks whether a short numerical description of the generator summarises the effects of changing it. Let R_A be its complete ordered transition table and L_A the logical bit length of an executable programme that reconstructs every entry. We call L_A *causal description length*, with *causal* referring to the known update mechanism. This is a length under a declared encoding, not the shortest possible programme. Its connection to Kolmogorov complexity, including the fixed decoder overhead, is given in Appendix F.

The distinction between the programme and its length matters. Executing the programme recovers the transitions needed to calculate cycles, frequencies and payoffs. Its length records how much space the encoded rules occupy, but does not identify those rules. We therefore examine whether equal lengths can accompany different payoffs and whether changes in length vary with changes in the long-run distribution.

BDM provides a comparison based on patterns in the output table without receiving the generating rules. Neither measurement enters the KL constraint or the payoff calculation. Their role here is to assess the limits of these summaries of the generator. Estimator conventions and the distinction between a complete mechanism and one output are retained in Appendix F.

C Network implementation and study design

C.1 An executable representation of the generator

The known Boolean rules are compiled into a shared ordered decision graph. Each decision tests one input coordinate and follows a zero or one branch; terminals supply output bits. Identical continuations are shared, following reduced ordered binary decision diagrams [4]. An ordered reference for each output makes the graph an executable description of the complete transition table.

The index-deconvolution framework also expresses the input positions that produce a one as decimal anchors plus additive offsets called *Sumandos*. These sets provide a way to inspect the graph and locate output patterns. The worked example below illustrates the representation; Appendix E gives the full unfolding, executable listings and bit count. Production compilation works directly from the known rules and need not enumerate the exported position sets.

C.2 A worked network with ten nodes

We use the existing baseline `ring_n10_s1000`, chosen by its fixed identifier for exposition. Figure 1 shows its connectivity and dynamics. The family name denotes the study generator; this example contains additional connections beyond a simple cycle. All gate parameter dictionaries are empty. MAJORITY is a strict majority, so its two-input use at node 3 requires both inputs to be one.

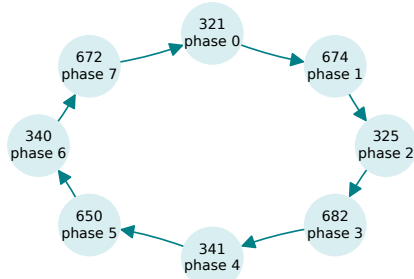
From output rows to cycles and frequencies. The same network has 1,024 input rows. Figure 3 selects rows that show how an output becomes the next input. These rows retain their original addresses; their display order follows the cycle so that the iteration is easy to read. For

example, input 65 produces state 674, which also appears as the output of input 321. State 65 is transient, whereas 321 and 674 belong to the same eight-state cycle.

A. One row is one update; the output address selects the next row

Address x	Input x0 ... x9	Output y0 ... y9	Next address F(x)	Cycle index
65	1000001000	0100010101	674	transient
321	1000001010	0100010101	674	G1, phase 0
674	0100010101	1010001010	325	G1, phase 1
325	1010001010	0101010101	682	G1, phase 2
682	0101010101	1010101010	341	G1, phase 3
341	1010101010	0101000101	650	G1, phase 4
650	0101000101	0010101010	340	G1, phase 5
340	0010101010	0000010101	672	G1, phase 6
672	0000010101	1000001010	321	G1, phase 7

B. Follow the actual eight-step cycle



C. Divide cycle mass among its phases

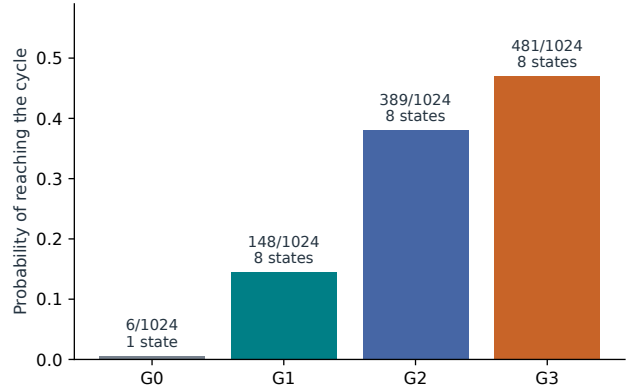


Figure 3: How does the repertoire produce a distribution? Panel A shows selected rows, each with its input address and next-state address. The extra cycle index records position in the trajectory, not position in the exhaustive table. Panel B follows the actual updates through cycle G1; phases 0–7 are labels and do not affect its probability. Panel C gives the probability of reaching each cycle from a uniformly chosen initial state. Dividing this mass by the number of cycle states gives each state’s long-run frequency.

Cycle	States	Basin size	Cycle probability	Each state
G0	1	6	3/512	3/512
G1	8	148	37/256	37/2048
G2	8	389	389/1024	389/8192
G3	8	481	481/1024	481/8192

Table 2: How much probability does each cycle receive? Basin size counts all initial states that eventually reach that cycle, including its own states. The four basin sizes sum to 1,024. The final column is the probability of each individual state on the cycle, not of the cycle as a whole.

Cycle G1 has a basin of 148 states and eight phases. It therefore receives mass 148/1024, and each phase has probability $148/(1024 \cdot 8) = 37/2048$. The fixed point 0 receives $6/1024 = 3/512$. Together the fixed point and three eight-state cycles give 25 states with positive long-run probability. There are 432 distinct one-step outputs, so counting those outputs once would give a different observation model.

A compact description of the same transitions. For node 0, the output is one when an odd number of input coordinates 5, 7 and 9 are one. One branch fixes coordinates 5 and 7 to zero

and coordinate 9 to one. Its decimal anchor is 512. The other seven coordinates are free; their additive offsets are the Sumandos, so this branch describes 128 input addresses without listing each one. Four such branches recover all 512 addresses at which this output is one. Appendix E shows the complete sets and a two-output pattern query.

Figure 4 shows the graph for output 0. It resembles a binary tree, but branches merge when they have the same remaining computation. The complete network has 19 reachable decision nodes, two shared terminals and ten ordered output references. Appendix H gives every decision record, so the example is not dependent on a schematic drawing.

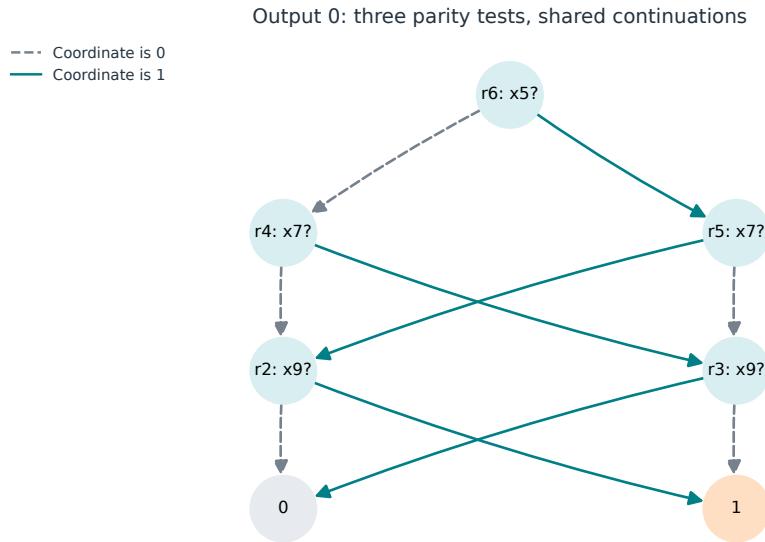


Figure 4: How can several paths share a short rule? Start at reference 6 and test the stated input coordinate. Follow the dashed branch for zero and the solid branch for one, until reaching a terminal. A terminal is the output bit, not a whole network state. Reused continuations keep this three-input parity rule to five decisions. The other output references belong to the same network programme.

The full graph has 19 decision nodes and uses 325 logical bits to reproduce the 10,240-bit transition table. This demonstrates compact reconstruction for the example. It does not establish an advantage over a direct encoding of the connections and gates, or a speed-up in computing the long-run distribution. For fixed N , the declared length formula depends only on the number of decision nodes: equal lengths can encode different rules. We examine the consequences of this loss of detail in the results.

C.3 Experimental design and verification

The first degree-five study used 240 base draws: four families, three sizes and 20 seeds. It produced 21,380 distinct main networks and 480 catalogues. The final replication kept the rules fixed and used 100 new seeds per family and size, giving 1,200 base draws. Sizes were $N = 8, 10, 12$; families were ring, hub, modular and sparse random. The ring and hub generators keep a structural template and vary gate assignments with seed. Consequently, 100 seeds do not mean 100 independently sampled graph structures.

Maximum indegree was five. Self-loops and empty input rows were excluded. When a generated node exceeded that limit, a fixed selection rule retained five input sources. The original matrix and the resulting changes were recorded. For every base, all one-edge additions and removals were enumerated in separate catalogues. Gates and parameters remained fixed during perturbation; the number of inputs and the validity of each gate were rechecked. Unchanged baseline entries remained in the audit and were excluded from perturbation effects and optima. When the same

complete network occurred more than once, it was evaluated once while retaining every catalogue membership.

The main gate assignments follow the frozen family generators. All twelve supported gates were additionally checked in a separate stress group: AND, OR, XOR, NAND, NOR, XNOR, NOT, IMPLIES, NIMPLIES, MAJORITY, KOFN and CANALISING. Pilot and stress results do not enter the main effect estimates. The replication main seeds were 1000–1099 and pilot seeds were 2000–2002.

Two payoffs were used. The target payoff is one at a deterministically chosen state with positive baseline probability and zero elsewhere. Its expectation is the frequency of that single target message type. The linear payoff is the fraction of active nodes in the state; its expectation rewards states with more ones. These are two explicit preferences over the same state alphabet, not measured probabilities of persuading a person. KL budgets were 0, 0.1, 0.25 and 0.5 nats, with 0.25 declared primary. Infinite KL values remained explicit. For each catalogue, the programme, BDM and dynamical changes were compared with its own baseline.

Each unique network received a serialised programme, full network data and an exact long-run distribution. Verification checked every output bit, serialisation round trips, an independent trajectory calculation, cycles, basins and rational probability mass. An independent Wolfram evaluation checked the ordered output digest. A digest is a reproducibility identifier; the reconstruction comparisons provide the underlying correctness evidence.

The replication initially stopped after a transaction of 48 Wolfram checks timed out. An explicit recovery amendment retained those failed records and repeated the 48 checks individually. Every replacement completed normally and matched its expected digest. The final audit replayed all 110,033 pilot and main networks with zero failures and checked 19,200 catalogue frontiers. It rechecked the historical Wolfram evidence; it did not rerun all Wolfram checks during this final replay. The release also passed 738 software tests and execution of the analysis notebook.

Programme and BDM changes are measured relative to each baseline and averaged within each base before comparing family, size and perturbation groups. These are descriptive associations, not a test of individual attack prediction. The estimator conventions, resampling procedure and comparison with the earlier study are given in Appendix G.

The explanatory extensions use the same saved ten-node baseline and its 74 edge additions. Cycle membership and basin sizes are independently checked by following all 1,024 initial states. Row BDM is a separate one-dimensional measurement described in Section F.3; it is not mixed with the replication’s matrix BDM results. The added complexity/payoff plots recompute both payoffs and forward KL from the saved exact probabilities. They introduce no new network sample.

D Additional catalogue and complexity results

D.1 Catalogue coverage and exact reconstruction

All 106,789 main networks completed the declared reconstruction and dynamical checks. They cover 2,400 complete nonempty catalogues and 105,642 changes relative to their baselines. Every programme reproduced its full transition table and was shorter than the raw table in this sample. Detailed length summaries are in Appendix G; the payoff results below use the exact distributions obtained from the transitions.

D.2 The answer under a structural constraint

For each completed catalogue, we selected the feasible change with greatest payoff. At the primary budget, some catalogues contained no feasible nonidentity change (Table 3). Keeping these cases visible is essential to describing the attacker’s actual choices.

Change	Payoff	Catalogues	Feasible	No feasible change
Add	Linear	1200	1186	14
Add	Target	1200	1186	14
Remove	Linear	1200	1103	97
Remove	Target	1200	1103	97

Table 3: Is there any allowed change below the alarm budget? Counts use $C = 0.25$ and all 1,200 bases per perturbation kind. Feasibility depends on the distributional constraint, so the two payoff definitions give the same feasibility counts. An unavailable optimum is not assigned payoff zero.

The ten-node example makes the distinction from the unrestricted answer concrete. It has 74 admissible nonidentity edge additions. At $C = 0.25$, 13 are feasible. For linear payoff, the selected change adds source 4 to target 8. Its expected payoff is exactly $51/64 = 0.796875$, with forward KL about 0.184507 nats. The saved unrestricted upper bound is 0.847502. For the target payoff in this same catalogue, the best attainable value remains 0.005859375 at each of the four budgets.

KL budget	Feasible changes	Best payoff	Saved upper bound
0.0	2	0.737500	0.737500
0.1	12	0.775000	0.811180
0.25	13	0.796875	0.847502
0.5	22	0.799219	0.883444

Table 4: What does the judge permit in the worked example? Each row uses the same ten-node baseline and all admissible edge additions. The best linear payoff is chosen after checking every candidate. The upper bound describes a distributional relaxation and may be unattainable by these edge changes.

The stored implementation evaluates a fixed logarithmic grid in the dual parameter, with explicit zero-budget and saturation cases. A grid value is a conservative upper bound, not a certified exact evaluation of the infimum in equation (8). The displayed gap can therefore include numerical slack as well as a structural restriction. All 19,200 recorded frontier values or unavailable cases passed the audit; each available optimum was below its recorded bound within the declared numerical tolerance.

D.3 Does description length summarise the payoff?

Figure 5 shows the same catalogue at the level of individual changes. Its horizontal axis is each changed network’s *actual* KL divergence, whereas Figure 2 uses the *allowed* budget and plots the best choice below it. Its colour is causal description length. Each point therefore answers three questions: how far do the frequencies move, what payoff results, and how long is the programme that generates the changed behaviour?

At $C = 0.25$, feasible programme lengths are 325, 339 and 353 bits. Among 339-bit programmes, linear payoff ranges from 0.737500 to 0.796875. The 353-bit feasible programme gives 0.725521. Thus length alone does not rank the payoff: different rules can require the same number of encoded bits yet produce different distributions. The selected source-4-to-target-8 addition uses 339 bits and achieves the best linear payoff at this budget. It has target-state payoff $1/256$, so improving the linear objective does not also optimise the target objective.

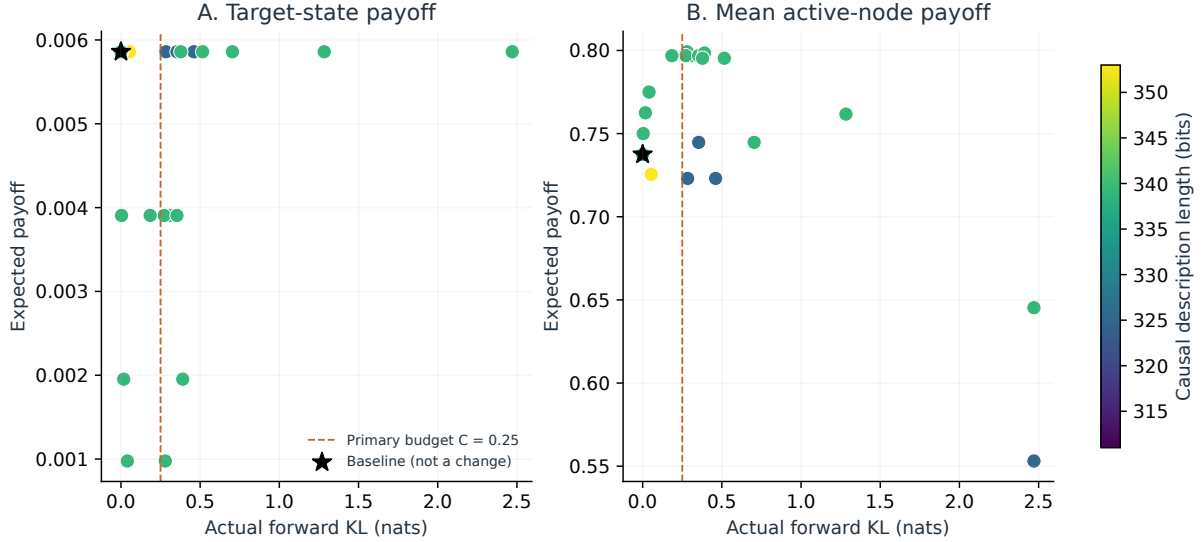


Figure 5: Do generators with similar description lengths have similar payoffs? Each coloured point is one admissible edge addition with finite forward KL; overlapping points may hide distinct changes. Of 74 changes, 29 have finite KL and 45 introduce positive long-run mass outside the baseline support, giving infinite KL. Those 45 cannot be placed on the finite axis and are retained in the accompanying data. The dashed line marks $C = 0.25$; 13 changes lie on its feasible side. The black star is the baseline for reference and is excluded from the changed-network optimum.

These examples show that description length alone is insufficient to determine payoff in this catalogue. The graph retains the specific rules that produce the distribution; the bit count discards that detail. Length measures a stored representation, not attack effort or persuasive power. There is no programme length assigned to the unrestricted bound, because its distribution need not be generated by a network in the catalogue. Appendix J retains the three-dimensional view of the same observations.

D.4 What the replication adds

Across every family/size group, mean programme length and mean matrix BDM increased under edge addition and decreased under removal. Their detailed responses were not interchangeable: correlations between base-mean programme and BDM changes ranged from -0.011 to 0.618 , while those between programme changes and total variation ranged from -0.432 to 0.327 . Appendix G gives the full comparisons.

These results describe associations between averages within each base. Total variation measures distributional change; the detector instead uses forward KL. The correlations therefore neither establish independence nor test whether complexity predicts detection of an individual change. Together with the worked example, they limit a universal interpretation of length as a summary of behaviour. They do not show that adding complexity improves the choice of an attack.

E Programme encoding and worked reconstruction

E.1 From index rules to an executable description

The method represents where output bits are one, then uses those positions to reconstruct each row. The index-deconvolution framework describes sets of row positions through decimal anchors and additive offsets called *Sumandos*. A compact schema fixes some bit coordinates and leaves

others free. If a stores the fixed one-bits and m stores the free coordinates, then

$$S(m) = \left\{ \sum_{i:m_i=1} \epsilon_i 2^i : \epsilon_i \in \{0,1\} \right\}, \quad P(a, m) = \{a + s : s \in S(m)\}, \quad a \& m = 0. \quad (9)$$

The decimal anchor records the fixed one-bits. The mask identifies the coordinates that are free to vary, and generates the additive offsets; it is not itself a list of Sumandos. Adding these offsets to an anchor unfolds the schema into input addresses. A union of schemas specifies the one positions of an output column; the remaining positions are zero. Reading all columns at the same address recovers the complete output row, including which node values occur together.

In this study the rules are known. The compiler turns them directly into an ordered decision graph, using node coordinates in ascending order. Each decision has a coordinate, a zero child and a one child. Identical branches are removed and identical continuations are shared, following reduced ordered binary decision diagrams [4]. Ten outputs use ten ordered references into the same graph. There is one programme for the network.

For teaching, the graph can be exported as decimal/free-mask schemas and rebuilt from them. Production compilation need not enumerate those schemas, their Sumandos, input assignments or possible output patterns. This distinction matters: a parity function may have many accepting paths but a short graph. The graph and the schemas describe the same input–output map. We measure the length of the executable graph encoding, including the information needed to decode it, rather than counting displayed anchors alone.

From a short rule to its addresses. For node 0, the output is one precisely when an odd number of coordinates 5, 7 and 9 are one. Its anchors are [512, 128, 32, 672]. Every anchor has free mask $351 = 1 + 2 + 4 + 8 + 16 + 64 + 256$, generating

$$S(351) = \{0, \dots, 31\} \cup \{64, \dots, 95\} \cup \{256, \dots, 287\} \cup \{320, \dots, 351\}. \quad (10)$$

Thus the decimal/Sumandos representation for this column is

$$[\text{decimals}, \text{Sumandos}] = [[512, 128, 32, 672], S(351)].$$

Each anchor contributes 128 positions. The four sets are disjoint and contain 512 positions in total. For instance, anchor 512 fixes coordinates 5 and 7 to zero and coordinate 9 to one. Adding each offset fills in the seven free coordinates.

This also supports a simple subpattern query. Node 1 copies input 0, so requiring $(y_0, y_1) = (1, 1)$ restricts the XOR one positions to addresses with $x_0 = 1$. On the branch above, the anchor becomes 513 and the free mask becomes 350. Across all four XOR branches, the full query is

$$\bigcup_{a \in \{513, 129, 33, 673\}} \{a + s : s \in S(350)\},$$

containing exactly 256 input addresses. The purpose is to locate which inputs generate a pattern. It does not change the network or introduce a new distribution by itself.

A. Keep one accepting branch of output 0

B. Read the same addresses in the output table

Fixed inputs: $x_5 = 0$, $x_7 = 0$, $x_9 = 1$

Decimal anchor: 512

Free coordinates: 0, 1, 2, 3, 4, 6, 8

Sumandos = $[0 \dots 31] \cup [64 \dots 95]$
 $\cup [256 \dots 287] \cup [320 \dots 351]$

Add each offset to 512: 128 addresses
 At every one of them, $y_0 = 1$.

Also require $y_1 = 1$: fix $x_0 = 1$.
 New anchor 513, free mask 350:
 64 addresses remain on this branch.

Offset	Address	Output $y_0 \dots y_9$	y_0	y_1
0	512	1000000000	1	0
1	513	1100010000	1	1
2	514	1010000000	1	0
3	515	1110010000	1	1
256	768	1000000001	1	0
257	769	1100010001	1	1
350	862	1010111101	1	0
351	863	1111101101	1	1

Selected rows; gaps are omitted.
 Orange rows satisfy $(y_0, y_1) = (1, 1)$.

Figure 6: How do decimals and Sumandos locate an output pattern? One accepting XOR branch fixes three input coordinates. Its anchor 512 plus 128 offsets identifies rows with $y_0 = 1$. Requiring $y_1 = 1$ also fixes $x_0 = 1$, leaving 64 rows on that branch. The displayed rows are selected examples, not the complete unfolded set; their output bits come from the verified network.

Output	Gate	Decimal anchors and free masks	One positions
0	XOR	(512, 351), (128, 351), (32, 351), (672, 351)	512
1	AND	(1, 1022)	512
2	XOR	(2, 1021)	512
3	MAJORITY	(5, 1018)	256
4	AND	(8, 1015)	512
5	XOR	(16, 1006), (1, 1006)	512
6	OR	(32, 983), (8, 1015)	768
7	XOR	(64, 959)	512
8	OR	(128, 895)	512
9	OR	(256, 703), (64, 959)	768

Table 5: Which positions are one in each output? Each pair is (decimal anchor, free-coordinate mask), rather than two explicit offset lists. Generate all submasks of the mask and add them to the anchor. The union within a row specifies one output column. All ten rows together specify the complete output table.

The following executable operations show both routes. The source network is saved with the manuscript evidence. The final assertion checks every one of the 10,240 output bits against the direct gate implementation.

```
from doppel_challenge.adapters import Network, repertoire
from doppel_challenge.repertoire_program import (
    compile_repertoire_program, compile_schema_program,
    export_output_schemata, serialize_program,
    deserialize_program, iter_output_rows)

# spec contains the saved connectivity, gates and parameters.
net = Network(n=10, C=spec['cm'], gates=spec['dyn'],
              params=spec['params'])
programme = compile_repertoire_program(net)
schemas = [export_output_schemata(programme, j)
            for j in range(10)]
payload = serialize_program(programme)
assert serialize_program(compile_schema_program(10, schemas)) == payload
restored = deserialize_program(payload, n=10)
rows = list(iter_output_rows(restored))
assert rows == repertoire(net)
print("Rows:", len(rows), "Output bits:", len(rows) * 10)
print("Output 0 schemas:", schemas[0])
```

Rows: 1024 Output bits: 10240

Output 0 schemas: [(512, 351), (128, 351), (32, 351), (672, 351)]

For an explicit unfolding of one schema, use:

```
anchor, mask = 512, 351
offsets, s = [], mask
while True:
    offsets.append(s)
    if s == 0:
        break
    s = (s - 1) & mask
positions = sorted(anchor + s for s in offsets)
assert len(positions) == 128
assert all(rows[x][0] == 1 for x in positions)
subpattern = [x for x in positions if rows[x][1] == 1]
assert len(subpattern) == 64
print("Sumandos:", sorted(offsets)[:4], "...",
      sorted(offsets)[-4:])
print("Positions:", positions[:4], "...", positions[-4:])
print("Branch rows with (y0,y1)=(1,1):", len(subpattern))
print("At input 513:", rows[513])
```

Sumandos: [0, 1, 2, 3] ... [348, 349, 350, 351]

Positions: [512, 513, 514, 515] ... [860, 861, 862, 863]

Branch rows with (y0,y1)=(1,1): 64

At input 513: [1, 1, 0, 0, 0, 1, 0, 0, 0, 0]

Repeating this construction for all ten columns recovers every transition. The resulting programme is a compressed representation of the original behaviour, not a newly inferred wiring diagram. Iterating those transitions then determines the cycles and the distribution used by the detector. This is the connection between the decimal/Sumandos representation, causal description length and the challenge.

E.2 Measuring the binary programme

We count the information needed to store and run the representation. For M decision nodes, the codec stores a self-delimiting count, M coordinate-and-child records, and N output references. Let $v = \lceil \log_2 N \rceil$ and $r = \lceil \log_2(M + 2) \rceil$. Its logical length is

$$L_A = 2\lceil \log_2(M + 1) \rceil + 1 + M(v + 2r) + Nr. \quad (11)$$

The count uses an Elias-gamma code for $M + 1$, which tells the decoder where that field ends without a separate length marker. Coordinate and reference fields use fixed-width, least-significant-bit-first representations. Decision identifiers follow a fixed order: children before their parent, zero before one, and outputs in column order. This removes arbitrary numbering differences, so equal reachable graphs under these conventions give equal serialisations.

For the example, $M = 19$, $v = 4$, $r = 5$. Table 6 shows every contribution. The programme occupies 41 bytes after adding three padding bits. The raw table is already binary: $10 \times 1024 = 10,240$ bits. The logical programme uses about 3.17% of that length. No input table, network topology, gate labels or parameters are supplied to the decoder.

Field	Logical bits
Node-count code	9
Decision records	266
Ordered output references	50
Total programme	325
Raw ordered output	10240
Byte padding (separate)	3

Table 6: What exactly is counted? The 325 logical bits contain enough structure to interpret the decisions and recover all ten outputs. The three padding bits only fill the last byte. The fixed interpreter overhead in equation (12) remains unmeasured.

Counting only bare anchors and offsets would measure a different object unless their grouping and output association were also supplied. The executable codec makes that decoding structure explicit. Compression is not guaranteed for every object: the same codec has a three-node AND/OR/XOR benchmark of 54 programme bits against 24 raw bits. A valid description may be longer than its data. No raw-table fallback is used.

E.3 What computation is saved?

Producing a compact rule and listing everything it generates are different computational tasks. The compiler can produce the exact description without writing the $N2^N$ -bit table. For the supported local gates, parity uses a number of construction states proportional to indegree d , and threshold rules use at most order d^2 states. Reading the current dense adjacency input already costs order N^2 . These observations explain why a compact representation is practical, while distinguishing that task from reading every output.

A query for one output row traverses the decisions needed by each output; in this direct local-gate setting it uses at most order Nd decisions. Printing the full table requires at least $N2^N$ bit writes. General Boolean composition, schema export and long-run inference do not inherit a universal linear-time guarantee. The present compiler also supports a specified gate library rather than arbitrary function minimisation. We make no numerical speed-up claim against another complete implementation in this paper.

Compilation is symbolic, but the experimental correctness checks, dynamics and BDM materialise

the complete small-network output tables. The distinction allows exact validation without treating the cost of validation as the cost of producing the compact description.

F Interpretation of the complexity measurements

F.1 Why measure causal description length?

Shannon’s information theory relates probabilities to average coding cost [2]. KL divergence measures the expected log probability ratio between two distributions. Here it supplies the constraint on changing frequencies. Its units are nats because the logarithm is natural.

Algorithmic information theory asks how short a programme can be while still generating a particular object [3]. Let R_A be the ordered matrix with row x equal to $F_A(x)$. We measure the logical binary length L_A of a programme that reconstructs R_A . Define $L_{\text{causal}}(A) := L_A$ under the stated compiler, codec and ordering conventions. This is the *causal description length* used here. We use *causal complexity* only with this operational meaning, not as a new expression for the exact Kolmogorov complexity. For a fixed universal interpreter U , this yields

$$K_U(R_A \mid N, \text{ordering}) \leq L_A + c_U, \quad (12)$$

where c_U includes the fixed decoder and interpreter overhead. Our codec is self-delimiting given N . The bound follows by supplying the valid programme to that decoder. The experiment measures L_A ; it does not measure c_U or find the shortest programme among all possible programmes.

Its value is to quantify how much explicit rule structure is needed by our representation to reproduce every transition. We can test that description by executing it. Comparing L_A with the raw table length measures compression; comparing $L_A - L_{A_0}$ across perturbations shows how this description changes when the mechanism changes.

The full transition table determines the trajectories and thus equation (6). The reverse need not hold. Different transition tables can have the same long-run distribution. Complexity of R_A and divergence of μ_A therefore describe different properties. A change in programme length alone does not determine whether the judge raises an alarm.

F.2 Why compare with BDM?

The Block Decomposition Method estimates algorithmic complexity by dividing an object into small blocks, assigning each distinct block a precomputed coding-theorem estimate, and accounting for repetition [5, 6]. In schematic form it uses $\sum_b [\text{CTM}(b) + \log_2 n_b]$, where n_b counts copies of block b . Our programme instead gives an explicit exact description built from the known Boolean rules. Comparing their responses to the same perturbations tests whether these two descriptions of regularity vary together. BDM is an estimate in its own units; it is not the stored length of our programme.

The comparison is useful because the two methods obtain their descriptions differently. Our method uses the known generating rules and provides an executable reconstruction. BDM examines patterns in the output object without receiving those rules. Agreement under perturbation suggests that some changes in the explicit description are also visible to the block estimator. Disagreement may reflect their different representations, block sizes or sensitivity to shared structure. Neither outcome measures accuracy against the unknown shortest programme.

Complexity is not needed to derive the solution in Section 2.1, and the specified detector never consults it. In the network experiment it is an additional description of the generator: how much rule structure accompanies a given payoff and distributional change? Treating complexity as an extra detection signal or an attacker cost would require a new model and a separate evaluation.

F.3 One output is not the whole mechanism

There are three different objects that an output-complexity measurement might refer to: one N -bit row $y = F_A(x)$, one node’s column across all 2^N inputs, or the complete $2^N \times N$ transition table R_A . The measurements must name which object they describe.

Write P_A for the network programme. It generates every row when given its input address. For a fixed interpreter and ordering, this gives

$$K_U(F_A(x) \mid x, N, \text{ordering}) \leq L_A + c_{\text{eval}}.$$

The same upper bound works for every x . It does not say that the shortest programme for each row has that length. For example, a row of ten zeros can be generated without reproducing the rest of the network. More generally, storing the ten output bits literally provides another description, with its own fixed decoding overhead.

If x is not supplied, a programme using the network must also specify which input to evaluate. Consequently,

$$K_U(F_A(x) \mid N, \text{ordering}) \leq L_A + K_U(x \mid N, \text{ordering}) + O(1).$$

If the network programme itself is supplied as auxiliary information, evaluating one row given x requires only a fixed evaluator: $K_U(F_A(x) \mid x, P_A, N, \text{ordering}) = O(1)$. These are different questions because different information is supplied.

Assigning L_A to every row is therefore meaningful as a label for the shared generating mechanism, not as a separate estimate of each row’s intrinsic complexity. Once that mechanism is transmitted, it can be reused for all inputs. Dividing its cost by the number of queries would describe an amortised communication cost, not the Kolmogorov complexity of a row. Similarly, an executable description of one output column could retain only the rules needed by that node. It need not cost as much as the whole network, and separately encoding all columns can duplicate shared rules.

BDM applied to each row instead asks about regularity within each short output string. Whole-table BDM also sees blocks spanning several rows and columns; summing row scores does not reproduce that measurement. For an illustrative check on the ten-node example, we additionally measure each output row with one-dimensional BDM using a single ten-bit block, without padding. At this size the score is the lookup estimate for that block [6]; it is distinct from the two-dimensional 4×4 convention used in the main experiment.

Input x	Output $F_A(x)$	Output bits	Row BDM	L_A (bits)
0	0	0000000000	22.784	325
65	674	0100010101	25.988	325
321	674	0100010101	25.988	325
512	1	1000000000	23.933	325
513	35	1100010000	26.676	325
1023	991	1111101111	24.578	325

Table 7: What remains fixed when output strings differ? Selected input addresses identify rows in the same ten-node network. Row BDM evaluates the displayed ten-bit string. The last column repeats the 325-bit network programme length as a mechanism label; it is not a measured minimum description length of that row. Inputs 65 and 321 give the same output and therefore the same row score.

Across all 1,024 rows, there are 432 distinct outputs and row BDM ranges from 22.784 to 29.428. The constant mechanism label and varying row scores show that one fixed generator can produce strings with different local regularities. These very short-string estimates do not establish differences in message meaning, persuasiveness or detection risk. Under a different

assignment of binary labels to message types, their values could change even when the outcome probabilities and payoffs are preserved.

G Supplementary complexity results

G.1 Estimator and comparison conventions

BDM used `pybdm` version 0.1.0 on the original output matrix, with non-overlapping 4×4 blocks and `PartitionIgnore` after zero padding. For $N = 10$, the 1024×10 matrix was padded to 1024×12 , adding 2,048 zero bits; $N = 8$ and $N = 12$ needed no padding. Padding is recorded as a BDM convention and is excluded from raw and programme lengths. For the worked example, BDM is 675.95 in BDM units.

Effects were averaged within each base and perturbation kind, then equally across bases in each family/size/kind group. The 95% intervals use 5,000 seeded resamples of base draws and describe sampling stability. They are not perturbation-level significance tests. The previous and final studies are compared group by group, without pooling them. Network-wide length summaries below are descriptive summaries of distinct stored networks.

G.2 The exact programme retains the full output table

All 106,789 main networks completed the declared checks. There were 2,400 complete nonempty catalogues of permitted changes and 105,642 changes relative to their baselines. The programme was shorter than raw for every main network in this sample. Table 8 reports the full range of programme lengths, including the longest cases. These are exact lengths under the codec conventions.

N	Networks	Raw bits	Programme bits: min / median / max	BDM median
8	21,204	2,048	65 / 257 / 439	609.94
10	34,832	10,240	131 / 339 / 695	733.52
12	50,753	49,152	199 / 419 / 787	1053.74

Table 8: How large are the complete output and its programme? Each row summarises distinct main networks of one size. The BDM median describes the original output matrix under the declared block and padding convention. It does not measure bits transmitted by the programme.

The ratios fall with N over this design because the exhaustive table grows exponentially while each local rule remains bounded in indegree. This result supports compact reconstruction for these networks. It does not establish compactness for arbitrary output tables or remove the cost of exhaustive long-run analysis.

G.3 Programme length, BDM and behaviour under perturbation

Mean causal description length and mean matrix BDM increased under addition and decreased under removal in every family/size group of the final replication. These are group means, not a statement that every single edge change has the same sign. Figure 8 places these effects alongside the change in the long-run distribution.

Agreement in average signs does not make the measures interchangeable. Across the 24 groups, Spearman correlations of base-mean programme changes with base-mean BDM changes range from -0.011 to 0.618 . Correlations between programme changes and total variation range from -0.432 to 0.327 . Thus the relation depends on family, size and perturbation kind. These

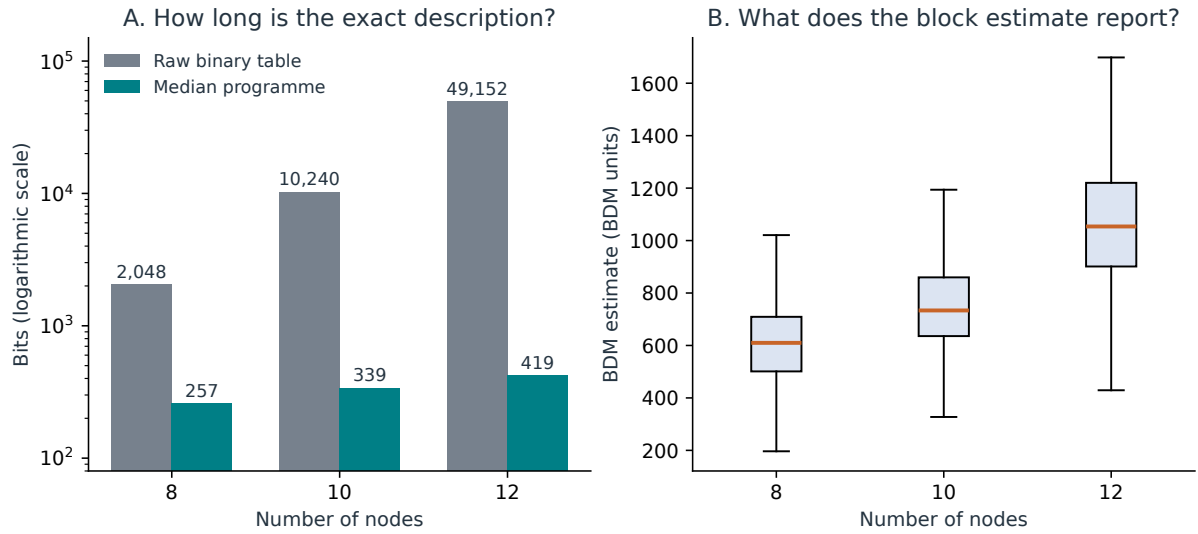


Figure 7: How much smaller is the exact generative description? Panel A compares raw length with median programme length on a logarithmic bit axis; labels give the actual values. Panel B shows BDM on its own axis: the box contains the middle 50%, the line is the median, and whiskers extend to the most extreme observations within 1.5 interquartile ranges. Outlier points are omitted from the drawing, not from the data or calculations.

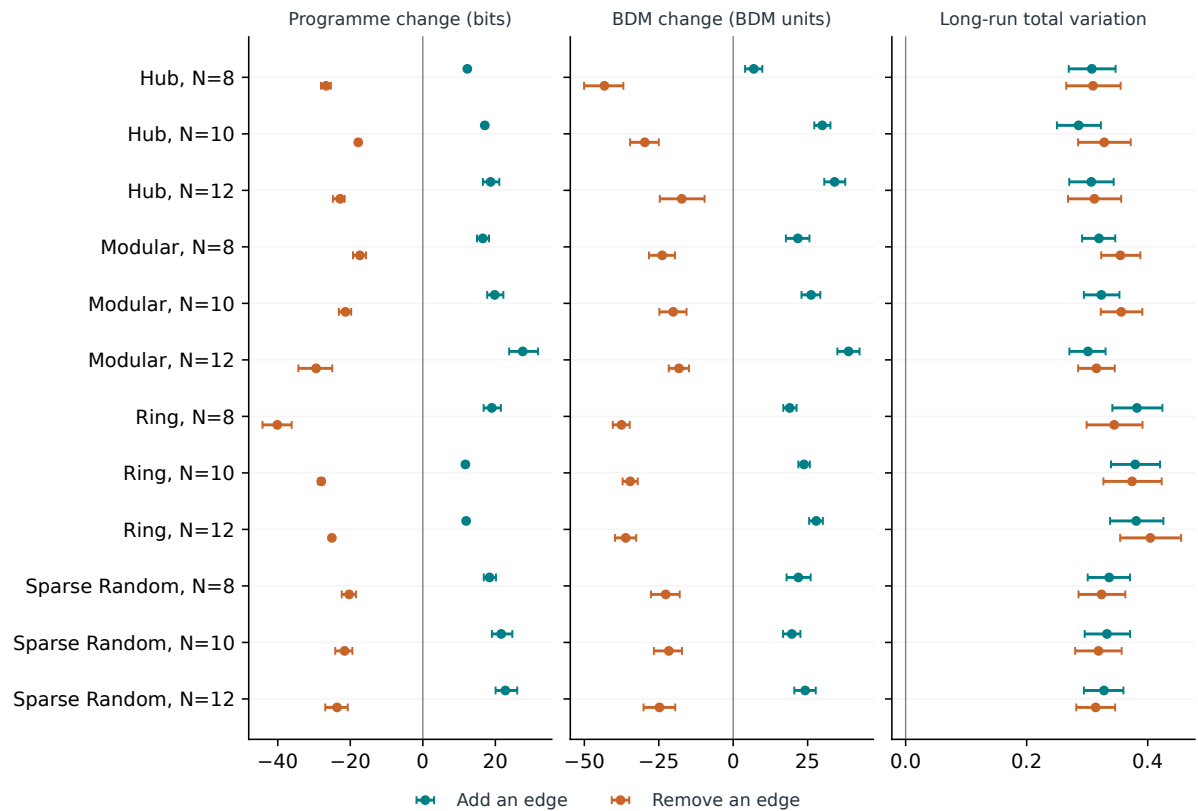


Figure 8: Do longer descriptions accompany larger behavioural changes? Read across each family and size. Teal points describe addition; orange points describe removal. Each point is an equally weighted mean of 100 base means, and each interval uses the declared base resampling procedure. The panels measure different quantities. In particular, total variation measures distributional change and has no sign corresponding to whether programme length increases or decreases.

descriptive associations do not establish a detection advantage or an accuracy ranking against an unknown Kolmogorov complexity.

G.4 Comparison with the earlier study

The final study increases base draws fivefold while retaining the same design. Table 9 reports the range of changes in the 24 matched group means. Complete group values are retained in the released comparison table; these ranges are descriptive differences and carry no cross-group significance claim. Appendix I gives all final group means.

Quantity	Smallest change	Largest change
Programme/raw (percentage points)	-0.182	1.043
Programme effect (bits)	-4.696	6.180
BDM effect (BDM units)	-8.154	11.260
Total variation effect	-0.019	0.082

Table 9: How much did the estimates change in the larger replication? Values are final minus previous group means. The smallest and largest differences may come from different groups. No perturbation rows are treated as independent replication units.

H The complete worked programme

References 0 and 1 are terminals. Each row below is a decision record. The output references for nodes 0 through 9 are

[6, 7, 8, 10, 11, 14, 16, 17, 18, 20].

Together with $N = 10$ and the stated ordering, these records define the complete output table. The canonical programme digest is

2271d1949e4e957370e8faddf94fed16d87b1cef5d0aaff60810f3aba22589b2

Reference	Coordinate	Zero child	One child
2	9	0	1
3	9	1	0
4	7	2	3
5	7	3	2
6	5	4	5
7	0	0	1
8	1	0	1
9	2	0	1
10	0	0	9
11	3	0	1
12	4	0	1
13	4	1	0
14	0	12	13
15	5	0	1
16	3	15	1
17	6	0	1
18	7	0	1
19	8	0	1
20	6	19	1

I All final group means

Programme/raw is the mean ratio after averaging perturbations within each base. ΔL is measured in bits; ΔBDM uses BDM units; TV is total variation of the long-run distributions. Additions and removals are separate. There are 100 base draws in each row.

Family	N	Change	$L/ R $ (%)	ΔL	ΔBDM	TV
hub	8	Add	14.38	12.28	6.91	0.308
hub	8	Remove	12.48	-26.68	-43.27	0.310
hub	10	Add	3.71	17.09	29.93	0.286
hub	10	Remove	3.37	-17.80	-29.70	0.328
hub	12	Add	0.93	18.70	34.07	0.307
hub	12	Remove	0.85	-22.80	-17.28	0.312
modular	8	Add	9.43	16.53	21.71	0.319
modular	8	Remove	7.78	-17.35	-23.85	0.355
modular	10	Add	3.34	19.81	26.18	0.323
modular	10	Remove	2.94	-21.34	-20.15	0.356
modular	12	Add	1.06	27.54	38.75	0.301
modular	12	Remove	0.95	-29.44	-18.21	0.315
ring	8	Add	11.80	19.02	18.99	0.382
ring	8	Remove	8.91	-40.12	-37.53	0.345
ring	10	Add	3.14	11.74	23.78	0.379
ring	10	Remove	2.75	-28.00	-34.59	0.374
ring	12	Add	0.77	11.94	27.90	0.381
ring	12	Remove	0.69	-25.08	-36.10	0.404
sparse random	8	Add	13.37	18.40	21.89	0.336
sparse random	8	Remove	11.48	-20.30	-22.66	0.324
sparse random	10	Add	3.79	21.61	19.69	0.333
sparse random	10	Remove	3.37	-21.57	-21.63	0.319
sparse random	12	Add	0.92	22.75	24.18	0.328
sparse random	12	Remove	0.82	-23.65	-24.77	0.314

J A three-dimensional view of the worked catalogue

Points at the same height share an encoded length, not necessarily a programme or a distribution. Points with the same payoff can also have different lengths. The third coordinate therefore describes the mechanism associated with a candidate choice; it does not change the detector's feasibility rule. A preference for shorter programmes among equally successful choices would be an additional optimisation criterion.

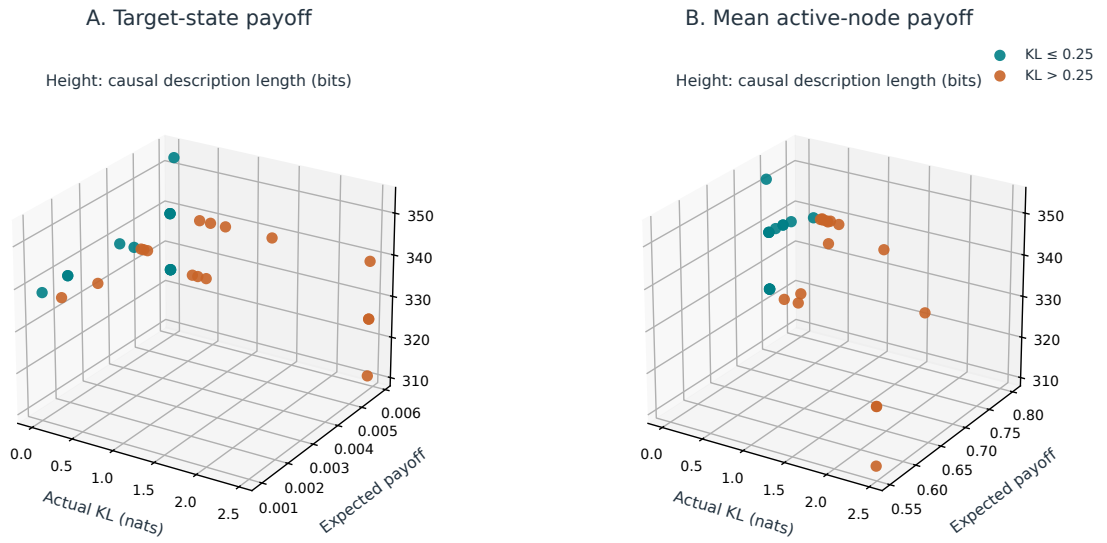


Figure 9: What does complexity add as a third coordinate? Height gives causal description length for the same 29 finite-KL changes shown in Figure 5. Teal points satisfy the primary budget; orange points exceed it. The 45 infinite-KL cases remain outside this finite plot. Each point is a complete changed network, not an individual message or output row. No surface is fitted between points.